



Research Article

The microstructure of layered ultramafic cumulates: Case study of the Bear Creek intrusion, Trinity ophiolite, California, USA



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ABSTRACT

In the Trinity ophiolite, California, USA, several mafic-ultramafic plutons intruded a peridotitic host 435 to 405 m. y. ago in a tectonic setting interpreted as an arc-related spreading centre. One of these intrusions, in the Bear Creek area, exposes basal ultramafic cumulates with igneous layering comprising an alternation of uncommonly thin (down to a few mm) layers of dunite, peridotite and pyroxenite that might be specific to this tectonic setting. These layers offer an excellent opportunity to characterise the microstructure of uncommon cumulates from the lower crust using EBSD (Electron Backscatter Diffraction). This “high-frequency” layering rests on underlying lherzolites and grades upward to more massive pyroxenites (i.e. clinopyroxenites with minor olivine-rich layers). Our field observations and data from the Bear Creek cumulates together with the preservation of magmatic features suggest the environment was tectonically stable after the emplacement of the cumulates. A detailed microstructural investigation of all minerals from the Bear Creek cumulates allows us to decipher their magmatic and plastic deformation history. In a structural reference frame defined by the compositional layering and the elongation direction of the surrounding host peridotites, olivine in the cumulates presents a [010]-fibre fabric and rarely a [001](010) fabric. Clinopyroxene shows a concentration of [010] axes normal to the layering plane with [100] and [001] defining girdles. Orthopyroxene mostly has a fabric with [100] and/or [010] subnormal to the layering plane and [001] scattered along a girdle in the plane of layering. All minerals show a strong fabric. We interpret the formation of the developed planar microstructures as a result of magmatic processes, with high contribution of crystal settling. To a lesser extent, compaction could have been operating and may be linked to the rare evidence of plastic deformation. Clusters of axes within the girdles of olivine and pyroxene CPOs preferentially appear close to the direction of elongation of the surrounding peridotites (i.e. N115°). EBSD analysis of the shape-preferred orientation of Bear Creek's minerals revealed a preferential alignment of the olivine and cpx long axis with the N°115 direction. This magmatic lineation and the preferred direction in the CPOs girdles are both consistent with the stretching lineation acquired during solid-state deformation by the mantle peridotite of the Trinity ophiolite. We suggest that a weak magma flux early on and/or an ongoing but limited regional stress could be responsible for these clusters. Although a direct coupling between asthenospheric flow and magmatic flow cannot be invoked in this context of melt intrusion in the lithosphere, this result highlights that the stress field applied on the mantle could have been still active and similar during the formation of Bear Creek intrusion. Our new field and microstructural data, together with previously presented petrological data, fit a scenario for the evolution of the Trinity ophiolite in which a mantle segment was intruded by a single large batch of primitive boninitic-andesitic melt. Our results emphasise the importance of considering the initial magmatic microstructures and the original shape anisotropy when investigating later deformation in ultramafic rocks.

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1. Introduction

In the past decades, pyroxenites have been the subject of numerous studies as their geochemical composition and microstructure provide important clues to petrological processes related to the development of heterogeneities in the Earth's upper mantle (e.g. Bodinier and

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Godard 2003; O'Reilly and Griffin 2013). Mineral fabrics in upper-mantle peridotites and pyroxenites are commonly interpreted as clues to solid-state ductile deformation that occurred by mechanisms such as dislocation and diffusion creep (e.g. Avé Lallemant and Carter 1970; Frets et al. 2012; Nicolas and Christensen 1987). When formed by dislocation creep, mineral fabrics will reflect the dominant slip system (or a combination of multiple slip systems each having a specific activity) that was active during the deformation episode (e.g. Ismail and Mainprice 1998; Karato 2008). The dependence of the activation of a slip system on parameters such as temperature, pressure, chemical composition or the presence of water in the mineral structure, which has been demonstrated by experimental studies, can be used to better characterise the deformation history of a rock and better constrain regional and global geodynamic models.

In contrast to pyroxenite layering in the mantle, the microstructures of pyroxene-rich intervals in lower-crustal mafic cumulates have received less attention from petrologists (e.g. Boorman et al. 2004; Feinberg et al. 2006; Higgins 1998, 2015; Williams et al. 2006). More specifically, the crystallographic preferred orientation (CPO) of pyroxenes in crustal pyroxenites has not yet been investigated using electron backscatter diffraction (EBSD) methods (e.g. Prior et al. 1999; Wright et al. 2011). Pyroxenitic cumulates are mostly present in the lower-crustal section of ophiolites and in layered magmatic intrusions (e.g. Benn and Allard 1989; Cawthorn and Walraven 1998; Ceuleneer and Le Sueur 2008; Clénet et al. 2010; Ying et al. 2013).

In magmatic intrusions, the rocks can record both magmatic and deformational processes that generate preferred grain orientation. The difficulty when studying this kind of lithologies is to differentiate between the two processes, which could be overprinted by one another. Previous studies reported that the microstructure of cumulates may result from a sequence of processes including crystal nucleation, crystal settling and the subsequent compaction of the mush (Cheadle and Gee 2017; McBirney and Nicolas 1997; Nicolas 1992). During magma flow, grains are aligned in a specific direction and will exhibit a planar-linear fabric on the rock, which then shows both a shape- and a crystallographic preferred orientation (CPO) (e.g. Boudreau and McBirney 1997; Charlier et al. 2015; Namur et al. 2015 and references therein; Vukmanovic et al. 2018; Williams et al. 2006). In this case, olivine crystals are generally elongated on [001] and show a correspondence between (010) and macroscopic markers like layering and igneous foliation (e.g. Benn and Allard 1989; Boudier 1991). A similar behaviour has been reported for the crystallographic axis of clinopyroxenes (cpx) with their [001] axes clustering along the magmatic lineation, which is itself contained in the magmatic foliation coinciding with the (010) plane (e.g. Benn and Allard 1989). During the subsequent magmatic history of the sample, the grains' SPO can be modified by late overgrowth that could also change the primary fabric (e.g. Brothers 1964; Cheadle and Gee 2017). With decreasing proportions of melt, the deformation regime approaches the solid state and possible grain boundary sliding assisted by diffusion-precipitation processes at low stress will take place (e.g. Paterson 2001). Finally, when the porosity becomes low enough, compaction occurs and tends to form a planar fabric; at this point the dominant deformation process becomes dislocation creep and leads to the formation of a CPO driven by plastic deformation.

Depending on the relative contribution of the processes involved, the respective strengths of the planar and linear component of the magmatic fabrics may vary. Consequently, the symmetry of the resulting fabric will range from an orthorhombic symmetry to a fibre symmetry (e.g. Benn and Allard 1989; Biedermann et al. 2016; Feinberg et al. 2006). The symmetry of a magmatic fabric can thus be used as an indicator of the magmatic-flow geometry and can be related to magma-chamber processes (e.g. Higgins 2015). Unfortunately, in natural samples, it is common that the initial magmatic fabric is overprinted by a later event of solid-state flow. This overprint results in a new fabric that carries different information from the original magmatic fabric (e.g. Satsukawa et al. 2013).

In the Klamath Mountains, California (USA), the Trinity ophiolite crops out as a large peridotite body intruded by a dozen ultramafic-mafic plutons of sizes from a few km to 10 km, exposed at different structural levels from the lower crust to the middle crust (Boudier et al. 1989; Cannat and Lécuyer 1991; Ceuleneer and Le Sueur 2008). Among them, the Bear Creek pluton exposes cumulates that crystallised in the vicinity of the mantle/crust boundary (Ceuleneer and Le Sueur 2008). As the Trinity ophiolite did not experience extensive regional metamorphism after its formation, it provides a unique opportunity to study the macro- and microstructure of pyroxene-rich cumulates and to observe magmatic and deformational processes occurring in magma chambers in a supra-subduction zone. In this work, field observations and EBSD data are used to characterise the microstructure and CPO of cumulate minerals from the Bear Creek pluton and its host peridotite. This constitutes the first EBSD study of crustal pyroxenites from a magmatic intrusion. These results are then used to discuss the interplay of magmatic and deformational processes in the development of mineral fabric in this context.

2. Geological background

The Trinity ophiolite is the largest (3000 km²) and best-preserved exposure of mantle mafic-ultramafic rocks cropping out in California. It is included in the Paleozoic Terranes accreted to the North American Cordillera during early Devonian times (e.g. Lindsley-Griffin 1977). Although its precise tectonic setting and emplacement history are poorly constrained, its association with arc-derived volcano-sedimentary formations together with geochemical and geochronological data suggest that it is a dismembered fragment of a former arc-related ocean basin that may have opened between Rodinia and Laurentia; the Trinity ophiolite itself could represent the youngest remnant of this oceanic domain (Brouxel et al. 1988; Coleman 1986; Mankinen et al. 2002; Wallin et al. 1988; Wallin and Metcalf 1998).

The Trinity ophiolite consists of an incomplete ophiolite sequence comprising mantle peridotites pockmarked by different plutons, which range from mafic-ultramafic to gabbroic and granodioritic in composition and several kilometres in size (Fig. 1A, Cannat and Lécuyer 1991; Ceuleneer and Le Sueur 2008). The mantle peridotites have a porphyroclastic texture that is typical of so-called mantle tectonites and that was developed under high temperature (close to the solidus) and low deviatoric stress, which have been attributed to asthenospheric flow (Le Sueur et al. 1984). An image of the flow geometry at the time it stopped was obtained by mapping the foliation and lineation imprinted by plastic deformation. The global survey of Le Sueur et al. (1984) revealed a rather organised pattern characterised by a sub-vertical foliation bearing a sub-horizontal stretching lineation, both trending NW-SE, i.e. oriented parallel to the inferred azimuth of the paleo-spreading axis. Local deviations from this regional pattern are frequent; some of them can be attributed to the development of late mylonitic shear zones.

The Trinity peridotites are mostly of Iherzolitic composition and resulted from a moderate degree of decompression melting which started in the spinel Iherzolite facies and ended in the plagioclase Iherzolite facies (Quick 1981). Patches of plagioclase Iherzolite represent about 10% of the surface of the Trinity peridotite (Quick 1981). Local concentrations of plagioclase (+/- cpx and spinel) attest to melt migration and segregation contemporaneous with the end of plastic deformation (Ceuleneer and Le Sueur 2008).

Abundant channels and dykes of pyroxenite (clinopyroxenite and cpx-rich websterite) crosscut the Trinity peridotites. The pyroxenites crystallised from a boninitic-andesitic melt (Mg / Si-rich and Ti-poor) derived from another mantle source that is not the Trinity Iherzolite as presently exposed (Ceuleneer and Le Sueur 2008). These boninitic-andesitic melts were in disequilibrium with the plagioclase Iherzolite at low pressure as attested by the melt/rock reaction features exposed. Harzburgites and dunites crop out at the margins of the pyroxenite

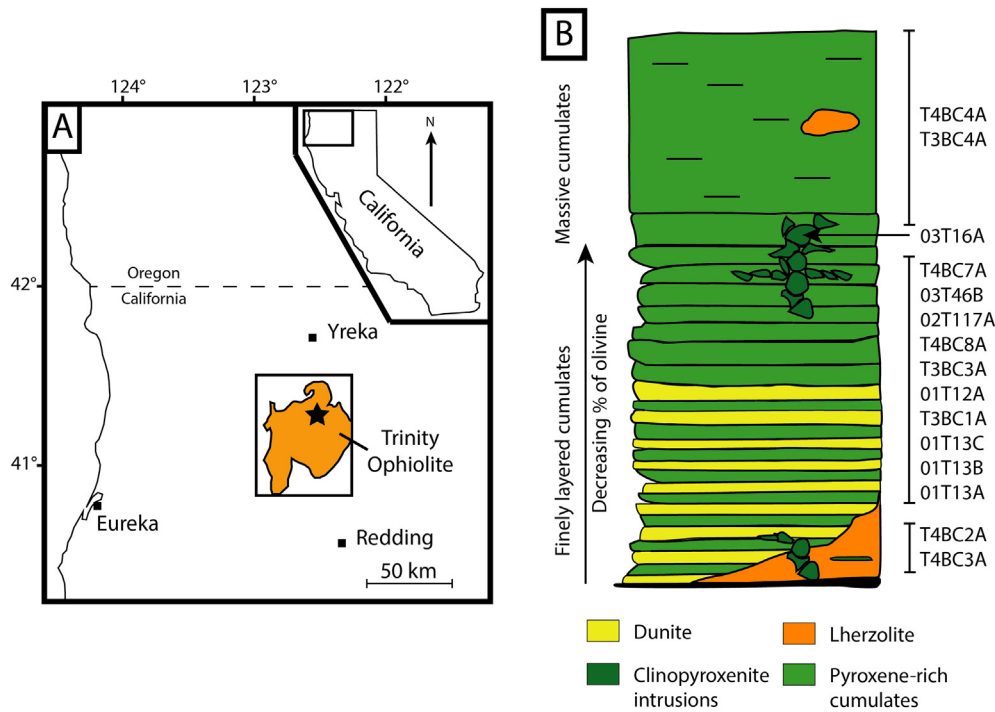


Fig. 1. Location of the Trinity ophiolite and schematic representation of the Bear Creek section modified after [Ceuleneer and Le Sueur 2008](#). (A) Location of the Trinity ophiolite in northern California, USA. The location of Bear Creek is marked by a star. (B) Schematic cross-section of the Bear Creek ultramafic complex. Basal Lherzolites are stratigraphically succeeded by the finely-layered cumulates. A decrease in bulk olivine content and an increase in grain size marks the transition to the massive cumulates. The total thickness of the cumulate section was reported to be ~500 m by [Ceuleneer and Le Sueur \(2008\)](#). Our sampling was performed on the first ~150 m. For more information about the geology of the Trinity ophiolite and the Bear Creek, please refer to [Le Sueur et al. 1984](#).

porous flow channels and dykes, to distances that can reach several meters ([Dygart et al. 2016](#); [Kelemen et al. 1992](#)). Reactive dunite with minor harzburgite also occurs as bodies a few tens of meters thick and hundreds of meters in lateral extent. They have been called “tabular dunites” by [Quick \(1982\)](#) in reference to their shape and their subhorizontal orientation. The large compositional diversity of the Trinity peridotite at various scales can be ascribed to the cryptic effects of the interaction between the moderately depleted mantle lherzolite and the ultra-depleted melt that crystallised as pyroxenite dykes ([Ceuleneer and Le Sueur 2008](#)). The overall tilt of the Trinity massif, at the regional scale, was found to be relatively moderate according to the orientation of the contacts between the different units making up the ophiolite and to the sub-vertical dip of the diabase dykes at upper-crustal levels ([Le Sueur et al. 1984](#)).

Among the plutons intrusive into the Trinity peridotite, granites and granodiorites were emplaced during Mesozoic times and are not related to the accretion history of the ophiolite. The mafic and ultramafic plutons are much older (405 to 435 Ma) ([Wallin et al. 1995](#); [Wallin and Metcalf 1998](#)) and their composition shows clear affinities with that of the pyroxenite dykes intruding into and reacting with the peridotite, which most likely belong to the same magmatic episode ([Ceuleneer and Le Sueur 2008](#)). They are differentiated, with a lithological succession consistent with the classical ophiolite igneous stratigraphy i.e., from the base to the top, primitive layered cumulates, more evolved cumulates devoid of developed layering and diabase dykes interpreted as the root of a sheeted dyke complex that is not preserved due to erosion. Contrasting with the classical ophiolitic model, these crustal units present a limited lateral extent (a few km at most) and were emplaced at various depths within the peridotite ([Cannat and Lécuyer 1991](#); [Wallin and Metcalf 1998](#)).

The Bear Creek area exposes the most complete and continuous section through one of these mafic and ultramafic plutons ([Ceuleneer and Le Sueur 2008](#)). Its lower contact with the mantle-derived rocks (i.e. of

lherzolitic composition) is affected by a late fault where the Trinity River flows ([Fig. 1B](#)). The lateral contact with the host rock is underlined by a screen of pegmatite made up essentially of large crystals of cpx (up to a few decimetres in length) partly replaced by magmatic hornblende and lower-temperature alteration minerals. In the pluton, the basal cumulate section is ~500 m thick and comprises an alternation of cm-thick layers of quite variable composition (dunite, wehrlite, pyroxenite, lherzolite...). Plagioclase is typically absent from this horizon; the modal abundance of olivine gradually decreases from bottom to top and it is virtually absent from the uppermost part of the cumulate pile. Plagioclase appears above this ultramafic sequence, in the upper half of the cumulate section, in a diffuse way at first (plagioclase pyroxenites), becoming increasingly abundant towards the top of the section. Its crystallisation always coincides with that of hornblende. The top of the cumulate sequence is characterised by abundant magmatic breccias crosscut by diabase dykes, and is not shown in [Fig. 1B](#) ([Le Sueur et al. 1984](#)).

[Ceuleneer and Le Sueur \(2008\)](#) attributed the aforementioned crystallisation sequence and the bottom-to-top evolution in modal and mineral compositions to the intrusion of a single batch of boninitic-andesitic melt (high MgO, silica and water content, low Al_2O_3 and low TiO_2) into the Trinity peridotites in a supra-subduction setting. Furthermore, the occurrence of diabase dykes at the top of the cumulate sequence were interpreted as representing the roots of a sheeted dyke complex and suggests that the magma injection occurred at relatively low pressure (<1 GPa) in a spreading setting ([Le Sueur et al. 1984](#)).

3. Field observations

In the Bear Creek plutonic section, the basal level (~500 m thick) is made up of serpentinised dunite layers alternating with pyroxene-rich and serpentine-free layers varying in composition between websterites,

wehrlites and lherzolites. The thickness of these layers ranges from several hundred micrometres (i.e. similar to the grain size) to several centimetres. We will refer to this compositional alternation between dunite and pyroxene-rich layers as “finely layered cumulates” (Figs. 2, 3A, B and Ceuleneer and Le Sueur 2008). The compositional layering is marked and is the unique macroscopic structural feature visible in the field. Its strike ranges from N115° to N85° with an associated dip of 30° to 40°S. The lateral extent of the layers exceeds several tens of metres and oblique structures made of clinopyroxenitic intrusions, generally with a coarser-grained texture, were observed. Peridotitic enclaves and pyroxene megacrysts (e.g. sample T3BC3A, Ceuleneer and Le Sueur 2008) also occur throughout this cumulate lower section, locally interrupting the compositional layering (i.e. at the hand specimen scale and thin section scale respectively; Supplementary Material B).

Towards higher stratigraphic levels, the compositional layering becomes increasingly difficult to observe due to the progressive

disappearance of olivine-rich layers (Figs. 1B and 3E). This results in seemingly more massive layers (up to ~250 m thick) that display a variable but always high cpx / orthopyroxene (opx) ratio and that we will refer to as “massive cumulates” (Fig. 3C and D). The strike of the layering, where still measurable, remains in the same range as that of the finely layered cumulates but its dip is more variable, ranging from 60 to 20°S.

The country rock in which the cumulates are embedded is of lherzolitic composition (Figs. 2A, 3E and F). The lherzolite is highly serpentinised and contains relatively fresher subparallel 1–5 cm thick pyroxenite veins highlighting a foliation which is also visible thanks to pyroxene grains (Fig. 3E). The foliation of the lherzolites ranges between N115° and N45° 30–40°S. In contrast to the cumulates, a mineral elongation was identified in the lherzolite and remains sub-horizontal. Fig. 2A shows the contact between the lherzolite and the finely layered cumulates which is unfortunately always hidden by loose blocks. A clear contact between the finely layered cumulates and the lherzolite foliation was only observed on a loose block, and was discordant. It is

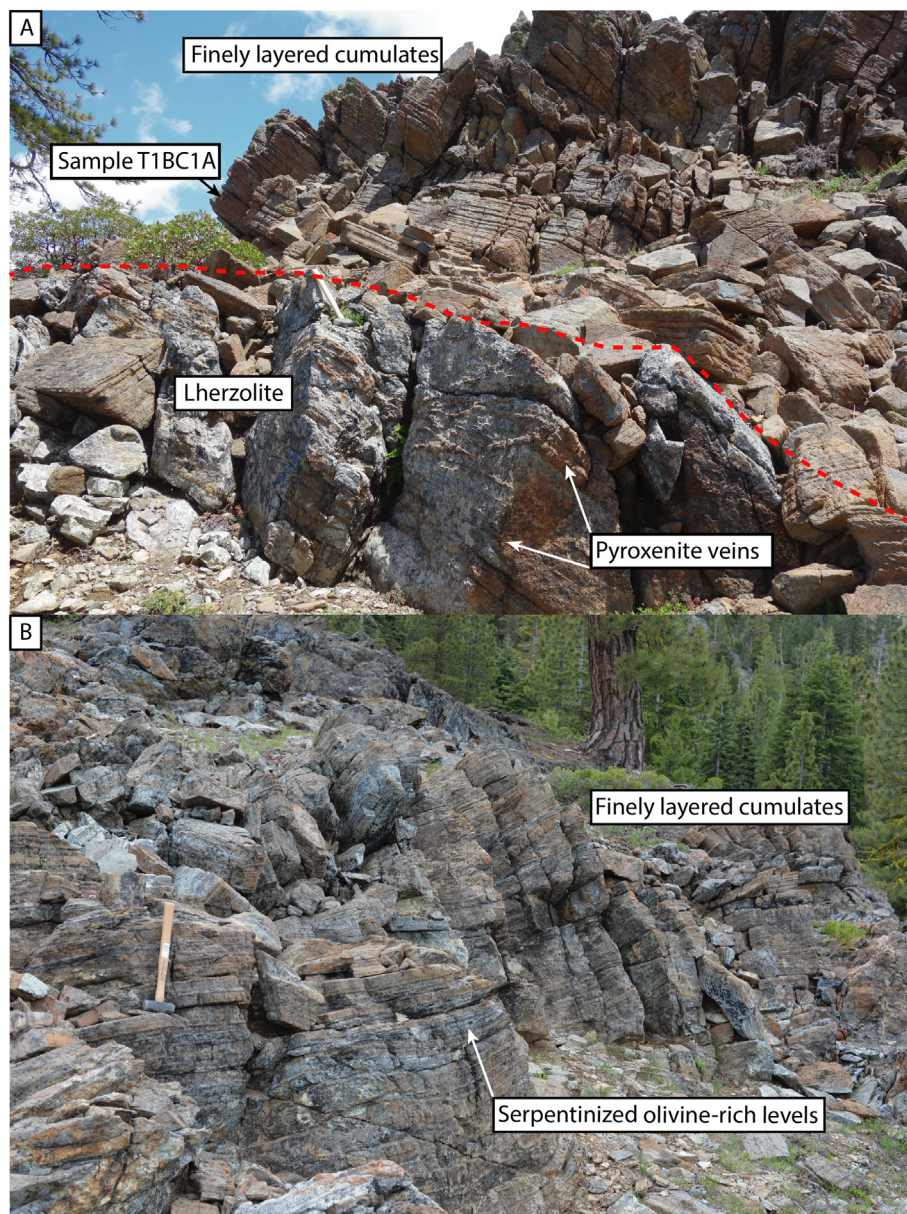


Fig. 2. Outcrops of finely-layered cumulates from Bear Creek. (A) Finely-layered cumulates structurally overlie basal lherzolites. Pyroxenite veins, both concordant and discordant to the lherzolite foliation, are visible in the lherzolites. Note the clear planar feature marked by the alternation of olivine-rich and olivine-poor layers. (B) Finely-layered cumulates displaying serpentinized olivine-rich levels. Scale is given by the hammer in both pictures.

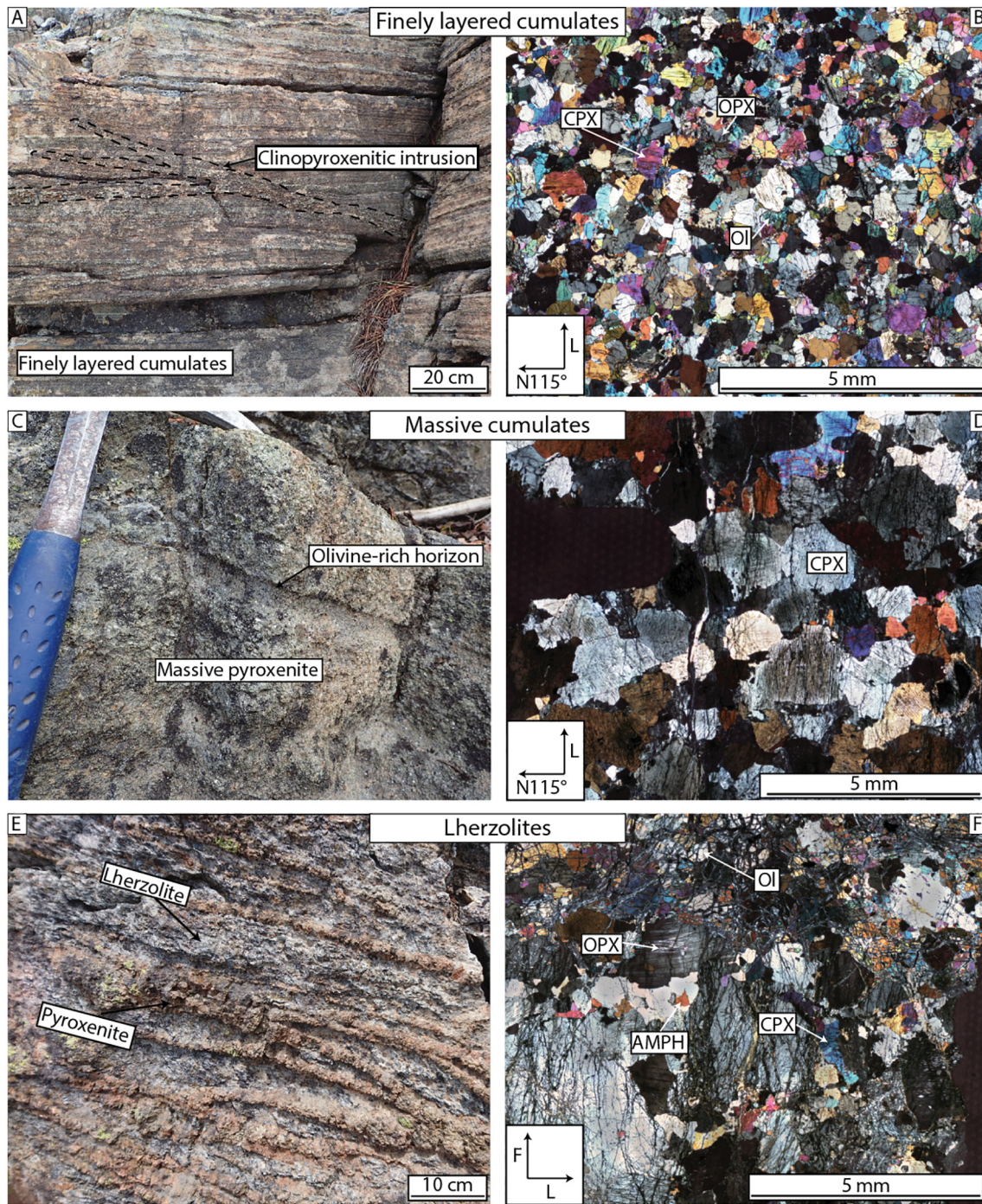


Fig. 3. Field pictures of the different Bear Creek lithologies. (A) Finely-layered cumulates with a clinopyroxenitic pegmatite intruding the magmatic layering; (B) Cross-polarised light photomicrograph of a finely-layered cumulate (T3BC1A); (C) Massive cumulates displaying a rare olivine-rich level; (D) Photomicrograph in cross-polarised light of a massive cumulate (T3BC4A); (E) Serpentinized lherzolite with pyroxenitic layers (F) Photomicrograph of a lherzolite (T4BC2A).

noteworthy that we did not observe any plagioclase within the Bear Creek finely layered cumulates or lherzolite.

In some locations along the section that we sampled, clinopyroxenite sills and dykes locally crosscut both the lherzolite foliation and the compositional layering of the cumulates (Fig. 3A and E). They appear to be dendritic extensions of meter-scale pockets of clinopyroxenitic intrusions that crop out locally in the lherzolite. The clinopyroxenitic dikes and sills can exhibit (1) a pegmatitic texture with idiomorphic, centimetric grains or (2) a finer-grained texture comparable to that of the massive cumulates. The two textures can coexist

within a single clinopyroxenite intrusion with the pegmatitic texture making up the core of the dyke and the finer-grained texture appearing at the rim.

4. Sample collection and analytical methods

4.1. Sample collection and preparation

Our sampling of the Bear Creek intrusion and its host rock covers a range of lithologies including the host mantle section (i.e. lherzolite)

and the finely layered and massive cumulates that crop out in the basal section of the intrusion. At Bear Creek, sampling was done along a section roughly perpendicular to the compositional layering of the cumulates (Fig. 1B and Table 1). Two lherzolites, ten finely layered pyroxene-rich cumulates, two massive pyroxenites and two clinopyroxenitic sills were selected for microstructural characterisation by EBSD. The samples are described in detail in Table 1 and Supplementary Material A, including GPS coordinates and microstructural data.

For samples in which no lineation could be observed in the field nor during sample preparation, thin sections were systematically cut in the plane normal to the foliation and containing the N115° direction. This direction corresponds to the azimuth of the regional stretching elongation in the mantle peridotites and to the strike of the intrusive diabase dykes, as inferred from the structural studies of Le Sueur et al. (1984) and Boudier et al. (1989). This direction was eventually used to relate the microstructure of the cumulate with that of the neighbouring peridotites. A suite of samples (i.e. 01T12A, 01T13A, 01T13B, 01T13C, 03T46B and 02T117A) collected in a previous field expedition was cut perpendicular to the compositional layering but was not oriented relative to the Trinity structural framework. For all samples, mineral fabrics were interpreted after checking the coincidence of the azimuth of the crystallographic-axis directions of at least two phases; petrofabrics were not interpreted where this criterion of internal consistency could not be verified.

4.2. Analytical methods

During sample preparation, a colloidal silica-water solution (mixed in a 70:30 proportion) was used for 5 min following mechanical polishing, in order to reduce surface damage. Thin sections were then coated with a ~ 6-nm-thick carbon layer to prevent charging during analysis. Crystallographic orientations of minerals were acquired using an HKL NordlysNano EBSD detector attached to a Zeiss EVO MA15 scanning electron microscope in the Geochemical Analysis Unit (GAU) at GEMOC/CCFS, Macquarie University, Australia. For all analyses, a 15-µm step size and a 20-kV acceleration voltage were used; working distance ranged between 15 and 20 mm. Automatic indexing was performed using the AZtec software (Oxford Instruments). The HKL software was used to perform a standard noise reduction and to extrapolate missing data with at least 8, 7, 6 and then 5 identical neighbours. Resulting EBSD data were then processed in MATLAB using the MTEX toolbox (Bachmann et al. 2010b, 2011). A value of 10° as the cut-off angle for the determination of the grains was used. Further

details on the data processing protocol in MTEX can be found in Henry et al. (2017) and Henry (2018).

To avoid oversampling of large grains, pole figures were plotted with one point per grain on a lower-hemisphere, equal-area stereographic projection. For strongly serpentinised samples (3 samples), pole figures were produced using every available measurements, which proved to yield results comparable to those of a standard one-point-per-grain model (Achenbach et al. 2011) (see Supplementary Material A for the number of grains and the type of representation). When twinning in cpx was detected during data processing, pole figures of cpx were drawn using all the available measurements.

Orientation density functions (ODF) have been calculated with a 10° Gaussian half-width for samples with >100 grains/individual measurements. The strength of the crystallographic preferred orientation was estimated using both the J- and M-indices (Bunge 1982; Skemer et al. 2005) and the appropriate ODF (i.e. calculated after all the measurements if twinning or serpentinisation occurred otherwise using one point per grain). In addition, the misorientation of each pixel was calculated relative to the mean orientation of its corresponding grain, and then combined as maps of misorientation relative to mean grain orientation. Any misorientation value greater than 10° away from the mean orientation was equated to 10° to allow for a clearer observation. Orientation tensors were also calculated for each crystallographic axis of each phase of interest using the method of Bachmann et al. (2010a). For each orientation tensor, eigenvalues for [100], [010] and [001] of olivine, cpx and opx were extracted and used to quantitatively describe the fabric symmetry using the BA- and LS-indices (Mainprice et al. 2015; Ulrich and Mainprice 2005). In order to quantify a possible shape-preferred orientation (SPO) of the grains that may not have been obvious in thin section, we used EBSD data to quantify their shape. To do so, we fitted an ellipse on every grain and extracted the angle between the long axis of that ellipse and the X direction (i.e. N115 direction in the layering plane). Results are then plotted as a multiple rose diagram with each data point being weighted by the surface area of the grain.

5. Petrography and microstructures of the Bear Creek intrusion and its country rocks

5.1. Lherzolitic country rock

The lherzolites are highly serpentinised, mostly preserving opx porphyroclasts (~ 2 mm), which are surrounded by fine-grained (~50 µm) olivine and minor cpx (Fig. 3F, samples T4BC2 and T4BC3). Undulose extinction, sub-grain boundaries and cpx exsolution along

Table 1
Lithology, texture and mineral abundance for the Bear Creek samples.

Sample	Lithology	Texture	Mineral abundance (%) ^a				
			Cpx	Opx	Olivine	Amphibole	Spinel
T4BC4A	Massive pyroxenite	Equigranular	95.66	0.74	2.98	0.61	0.01
T3BC4A	Massive pyroxenite	Equigranular	88.59	1.16	10.15	0.08	0.02
03T16A	Clinopyroxenitic intrusion	Pegmatitic	86.33	6.03	5.61	1.95	0.08
T4BC7B	Finely layered cumulates	Equigranular	89.2	0.56	6.1	4.12	0.02
03T46B	Finely layered cumulates	Equigranular	17.44	49.67	31.62	0.77	0.5
02T117A-coarse	Clinopyroxenitic intrusion	Pegmatitic	99.67	0.25	0	0.07	0.01
02T117A-cumulates	Finely layered cumulates	Equigranular	35.4	37.63	26.57	0.35	0.05
T3BC3A	Finely layered cumulates	Equigranular	58.44	26.61	12.56	2.37	0.02
01T13C	Finely layered cumulates	Equigranular	25.18	30.89	39.31	3.58	1.04
01T13B	Finely layered cumulates	Equigranular	81.13	6.01	8.87	3.88	0.11
01T13A	Finely layered cumulates	Equigranular	54.33	19.29	11.42	14.88	0.08
T3BC1A	Finely layered cumulates	Equigranular	39.66	21.52	34.92	3.83	0.07
T4BC8A	Finely layered cumulates	Equigranular	23.16	56.2	20.03	0.58	0.03
01T12A	Finely layered cumulates	Equigranular	26.14	44.3	25.58	3.53	0.45
T4BC2A	Lherzolite	Porphyroclastic	25.44	25.83	44.92	2.27	1.54
T4BC3A	Lherzolite	Porphyroclastic	5.59	30.69	63.56	0.02	0.14

^a Calculated on indexed EBSD data.

cleavages are observed in the large grains. Olivine is coarse-grained (~3 mm) away from opx-rich aggregates, and free from observable deformation markers. No obvious shape-preferred orientation of the grains is visible in thin section.

5.2. Finely layered cumulates

In some of our samples (e.g. T3BC1A, Figs. 3B, 4A and Table 1), olivine, opx and cpx are relatively homogeneously distributed. However, in other samples high-frequency alternation between olivine-rich and pyroxene-rich millimetre-thick layers is visible (Fig. 4B and C). The transition between layers is sharp and no gradual change in the mode of

olivine is seen. Mineral modes can vary strongly within each compositional layer, within the sample and from one sample to the others (Fig. 4 and Ceuleneer and Le Sueur 2008). Grain-size distribution is rather homogeneous with a mean of ~400 μm for all phases. Grain boundaries between phases are curved/lobate and 120° contacts are common. Grain boundaries can be either straight or curvilinear between grains of same phase and 120° triple junctions are common. We did not preferentially observe either convex or concave grain boundaries of a specific phase at contacts between two phases. Some olivine and cpx grains can be elongated on the layering plane but no clear shape-preferred orientation is seen for any phase. Only rare deformation markers can be observed in the finely layered cumulates (i.e. straight

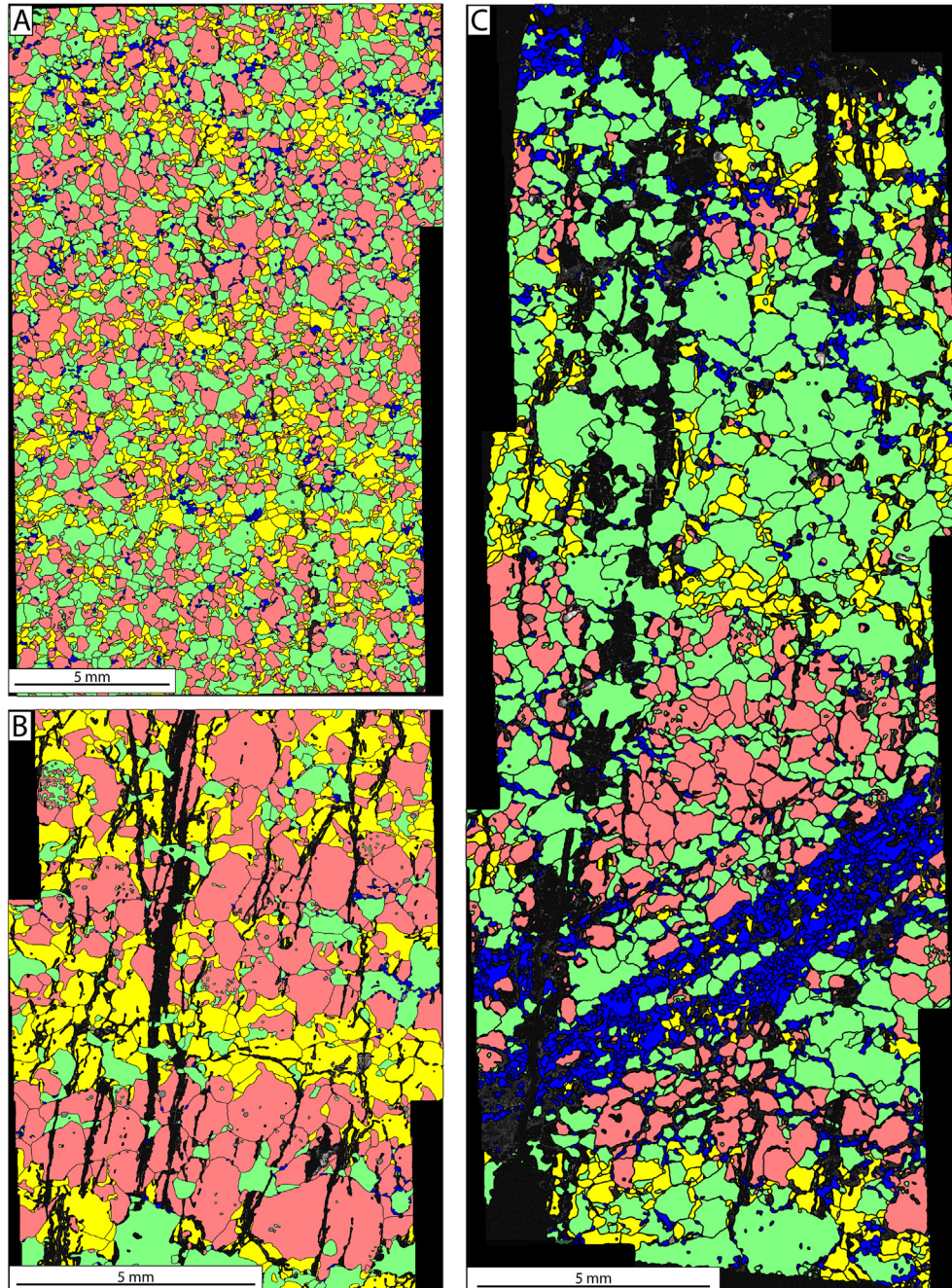


Fig. 4. EBSD-derived phase map showing the range of microstructures observed in the finely-layered cumulates. Cpx is pictured in green, opx in red, olivine in yellow and amphibole in blue. (A) relatively homogeneous finely-layered microstructure of sample T3BC1A; (B) microstructure of sample 03T46B displaying a strong compositional layering with opx-rich layers alternating with olivine-rich layers; (C) microstructure of sample 01T13A showing an amphibole vein crosscutting the layering. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

sub-grain boundaries in olivine, and to a lesser extent in cpx and opx; Fig. 5A). Cpx often presents twins with continuous to stepped twin-boundaries crosscutting the whole grain in most cases. At the thin section scale, macrocrysts of opx (~1 cm) can bend the layer contacts and can cross over multiple layers (e.g. T3BC3A, Supplementary Material B). Amphibole appears as small interstitial grains (~150 μm), often associated with cpx. An amphibole-rich vein crosscutting the cumulates was observed in sample 01T13A (Fig. 4C).

5.3. Massive cumulates

In contrast to the finely layered cumulates, this lithology is dominated by coarse-grained minerals (~500 μm to 1 mm) and is made up essentially of cpx alternating with rare levels of olivine with minor opx (i.e. compositional layering visible in the field, Fig. 3E). Cpx grain boundaries are mostly lobate, but can also be locally straight (i.e. Fig. 5B). In the finer-grained fraction (~50 μm grain size, <5% of the sample surface area), olivine and opx assemblages appear with mainly lobate/curved grain boundaries. Triple junctions are visible between most cpx grains and indicate a local textural equilibrium. Internal deformation in the form of subgrains and undulose extinction is observed in olivine, and sometimes in cpx grains. EBSD-derived maps reveal that concentric misorientation is present in some cpx grains (Fig. 5B). As in the finely layered cumulates, cpx macrocrysts (~1 cm) locally occur and twinned cpx are also observed (sample T4BC4A, Supplementary Material B).

5.4. Shape-preferred orientation of the Bear Creek cumulates

The shape-preferred orientation (SPO) of the grains in the Bear Creek cumulates was investigated by EBSD mapping. In the finely layered cumulates, our results show a clear elongation of the cpx and olivine grains in the layering plane (Fig. 6A). Opx present an SPO with a preferred direction of elongation normal to the layering plane.

In the massive cumulates, the quantification of cpx SPO shows a range of orientations from sub-parallel to olivine and cpx in the finely layered cumulate (sub-parallel to N115 direction) to a maximum at a small angle to the lineation (Fig. 6B).

5.5. Clinopyroxenite intrusions

Clinopyroxenite intrusions are composed dominantly of sub-idiomorphic to idiomorphic coarse-grained cpx (ranging in size from ~5 mm to ~2 cm, Fig. 7) with interstitial assemblages of olivine, cpx, opx and hornblende (1–5 mm). Grain boundaries appear linear and 120° triple junctions are common among cpx grains. Rare macrocrysts of euhedral to subhedral cpx locally exhibit twinning with straight to stepped twin boundaries (Fig. 7B, C and Ceuleneer and Le Sueur 2008). In contrast, olivine and opx are xenomorphic, with mostly lobate or curvilinear grain boundaries; triple junctions and deformation markers are absent. Locally, these intrusions grade into finer-grained layers (e.g. sample 02T117-A) where their texture becomes similar to that of the massive cumulates (Figs. 7A and 8A).

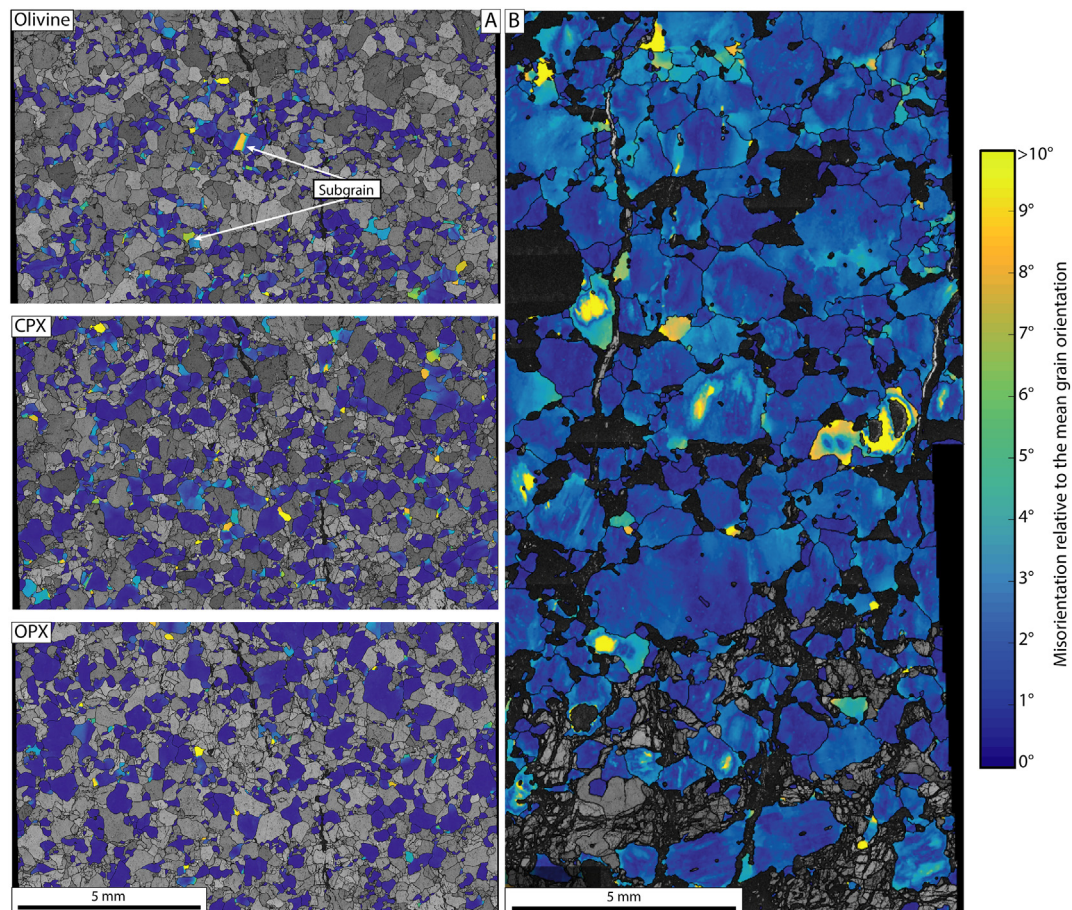


Fig. 5. Maps of misorientation relative to the mean orientation of the grains for (A) olivine, cpx and opx of sample T3BC1A, a finely-layered cumulate and for (B) cpx from sample T3BC4A, a massive cumulate.

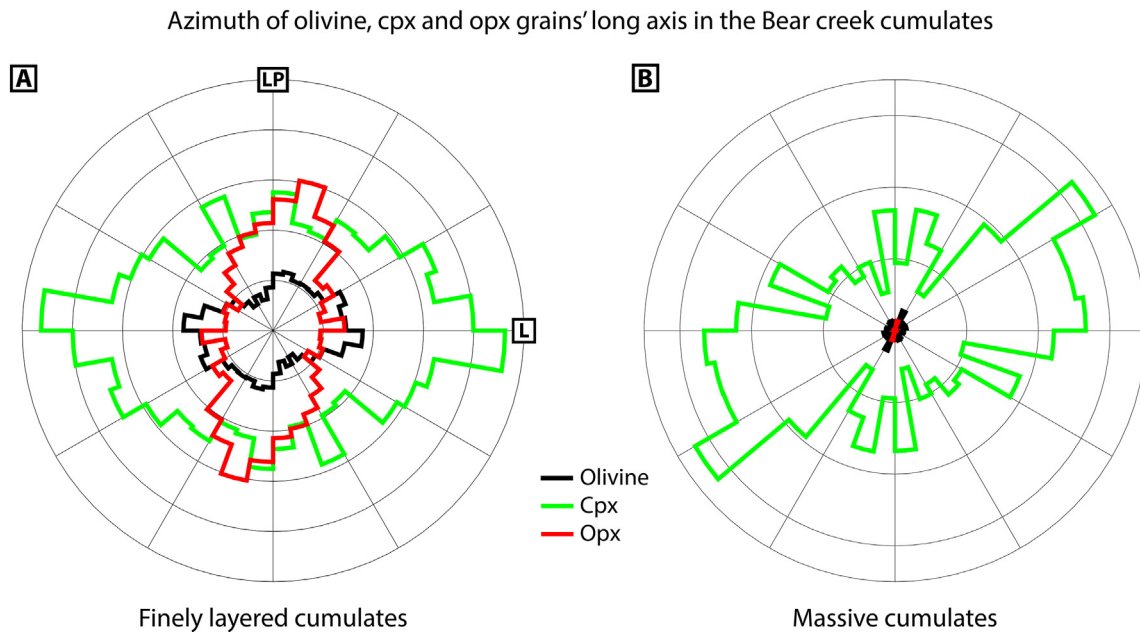


Fig. 6. Azimuth of the long axes of olivine, cpx and opx from the Bear Creek finely layered cumulates (A) and massive cumulates (B). Data are weighted by each grain's surface area and are plotted in the structural framework internal to the samples. LP and L respectively represent the pole of the layering plane (i.e. compositional layering) and the N115° direction in which the sample were prepared.

6. EBSD results

6.1. Lherzolitic country rock

The mineral fabrics of olivine, cpx and opx from the two analysed lherzolites are presented in Fig. 9. The olivine fabric shows a strong concentration of [010] axes perpendicular to the foliation plane, while either [100] or [001] axes are sub-parallel with the lineation direction depending on the sample. The cpx fabric has [001] axes parallel to the lineation and [010] axes perpendicular to the foliation plane. Opx exhibits a cluster of [001] axes around the lineation direction and a weak point maximum for [100] and [010] axes, both fitting the foliation pole (Fig. 9). Amphibole, when present, has a fabric mimicking that of cpx (Supplementary Material D). The fabric symmetry for olivine and opx in the lherzolite plots within the orthorhombic domain while cpx is spread between the orthorhombic and planar domains (Fig. 10).

6.2. Finely layered cumulates

The finely layered cumulates were investigated using nine samples; results are shown in Fig. 11 (see also Figs. 4, 5A and Table 1). The grain sizes calculated using EBSD data do not show any variation across this part of the section (Supplementary Material C). The olivine aspect ratio is always slightly higher than that of the pyroxenes, for a smaller grain size. The olivine fabric is dominated by a strong concentration of [010] axes perpendicular to the layering plane. Olivine [100] and [001] axes are distributed as girdles in the layering plane (Fig. 10). This observation is consistent with the BA-indices (Fig. 10), suggesting [010]-axial symmetries for the olivine fabrics. Other samples exhibit clustering of the [001] axes (e.g. samples T4BC7B, 01T12A) or the [100] axes (e.g. samples T3BC3A and 01T13A, Fig. 11) around the N115° direction. Cpx displays a clustering of [010] axes, and to a lesser extent of [100], normal to the layering plane. The cpx [001] axes mostly form girdles in the layering plane, as indicated by their low LS-indices (Fig. 10, Sup. Material A), but may also define a sub-maximum within the girdle, close to the N115° direction. Cpx fabrics are dominated by a planar symmetry, as again indicated by their low LS-indices (Fig. 10). The opx fabric usually has a weak concentration of both [100] and [010] axes next to the

layering pole, while the [001] axis forms a girdle with a sub-maximum aligned in the N115° direction (Fig. 11). Orientation maps reveal that two evenly-spread populations of opx may coexist and are responsible for the [100] and [010] clusters near the pole of the layering plane (Supplementary Material B). Opx appears to have higher BA-indices than LS-indices, suggesting a greater contribution of [010] and [001] of opx to the planar structure (Figs. 10 and 11). When amphibole is present, its fabric is similar to that of cpx (Fig. 10). As described earlier, a thin amphibole vein crosscutting the magmatic layering is observed in one sample (01T13A, Fig. 4C). The fabric of the amphibole making up the vein is similar to that of the amphibole in the finely layered cumulates that appears as interstitial grains between olivine, opx and cpx grains. Twinning of the cpx in the finely layered cumulates is characterised by EBSD and represents a rotation of about 180° around [001].

6.3. Massive cumulates

Two samples (T3BC4A, T4BC4A) were used to investigate the fabric of massive cumulates (Fig. 12). In both samples, the cpx [010] axes form a cluster normal to the layering plane while [001] axes are scattered and form a girdle in the layering plane (Fig. 12). In sample T3BC4A, olivine yields a fabric with the [001] axes clustering weakly along the N115° direction and the [010] axis normal to the layering. The number of olivine grains in sample T4BC4A does not allow for a clear interpretation of the fabric. In terms of fabric symmetry, cpx in our two samples has a low LS-index, suggesting a strong planar component for the fabric, while the only occurrence of olivine plots in the orthorhombic domain (Fig. 10). In the massive cumulates, the occurrence of opx and amphibole is so low that no reliable observation of the mineral fabric can be made.

6.4. Clinopyroxenitic intrusions

The pegmatitic microstructure in the clinopyroxenitic intrusions was investigated using sample 03T16A (Fig. 7B and C, Supplementary Material E). The determination of fabrics is hampered by the lack of structural framework inherent to this lithology; internal deformation is lacking, and the large grain-size limits the number that can be

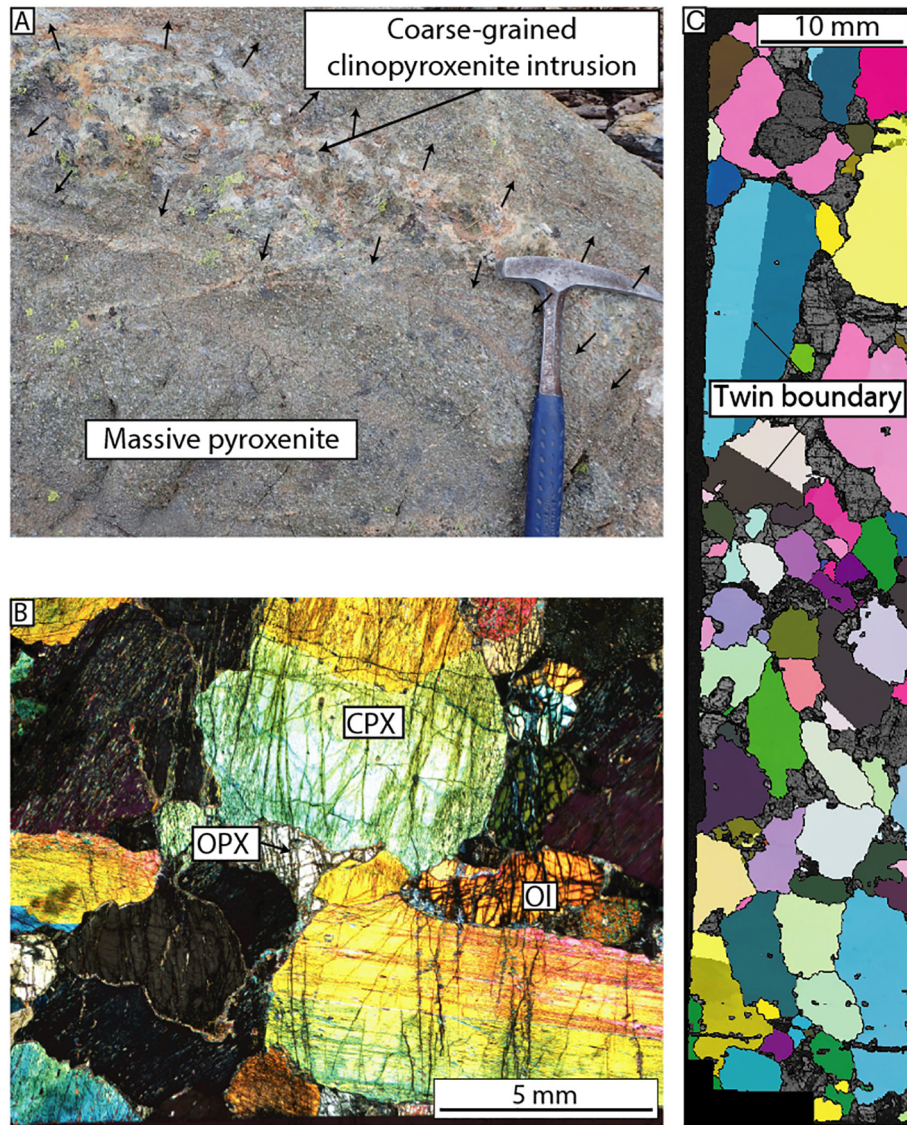


Fig. 7. (A) coarse-grained clinopyroxenitic intrusion in massive cumulates with a diffuse contact between the two lithologies. Black arrows indicate the textural transition between the pegmatitic texture and the massive cumulate texture; (B) Photomicrograph in cross -polarised light of the large grains in a clinopyroxenitic intrusion (03T16A); (C) Orientation map for cpx in sample 03T16A highlighting the numerous cpx twins in the pegmatitic portions on the clinopyroxenitic intrusions.

analysed. In sample 03T16A, cpx and amphibole are the only phases with sufficient grains to calculate an ODF. The cpx [001] axes exhibit point maxima for sample 03T16A (Supplementary Material E). In sample 03T16A, amphibole displays a fabric similar to that for cpx.

The microstructure of the finer-grained section of the clinopyroxenite intrusion was studied in sample 02T117-A, which represents the contact between a clinopyroxenitic intrusion and the finely layered cumulates (Fig. 8A, B). In the clinopyroxenite part, only cpx was observed while cpx, opx and olivine are all visible in the finely layered cumulates. The cpx in the clinopyroxenite intrusion shows [010] normal to the intrusion plane with [001] axes forming a girdle on that plane. The fabrics and textures observed in the finely layered cumulate are comparable to that previously described. Similarly, the amphibole fabric matches that of cpx (Supplementary Material E).

7. Discussion

Field observations, textural and microstructural data are integrated from the bottom to the top of the Bear Creek intrusion and its associated mantle domain in order to unravel the solid state and magmatic

processes that took place in the Bear Creek. Implications of the deformation episodes and the related conditions are then considered in the context of the evolution of the Trinity ophiolite.

7.1. Conditions of deformation in the host Trinity mantle

In this section, we first discuss the EBSD data in the light of the field and microstructural observations and then use them to estimate the conditions under which plastic deformation occurred prior to the intrusion.

Iherzolitic cpx displays a SL-type fabric ([010] and [001] axes clustering along the foliation pole and the lineation direction respectively), suggesting deformation at high temperature, and a combination of pure and simple shear (Bascou et al. 2002; Helmstaedt et al. 1972). Opx displays a ABC fabric (i.e. [001] axis along the lineation direction and weak cluster of [100] and [010] normal to the layering, Jung et al. 2010) while olivine shows an A-type fabric (T4BC2A), consistent with deformation of the Iherzolite by dislocation creep at high temperature and low stress under “asthenospheric” conditions in the upper mantle (Jung and Karato 2001; Michibayashi and Oohara 2013). Sample T4BC3A

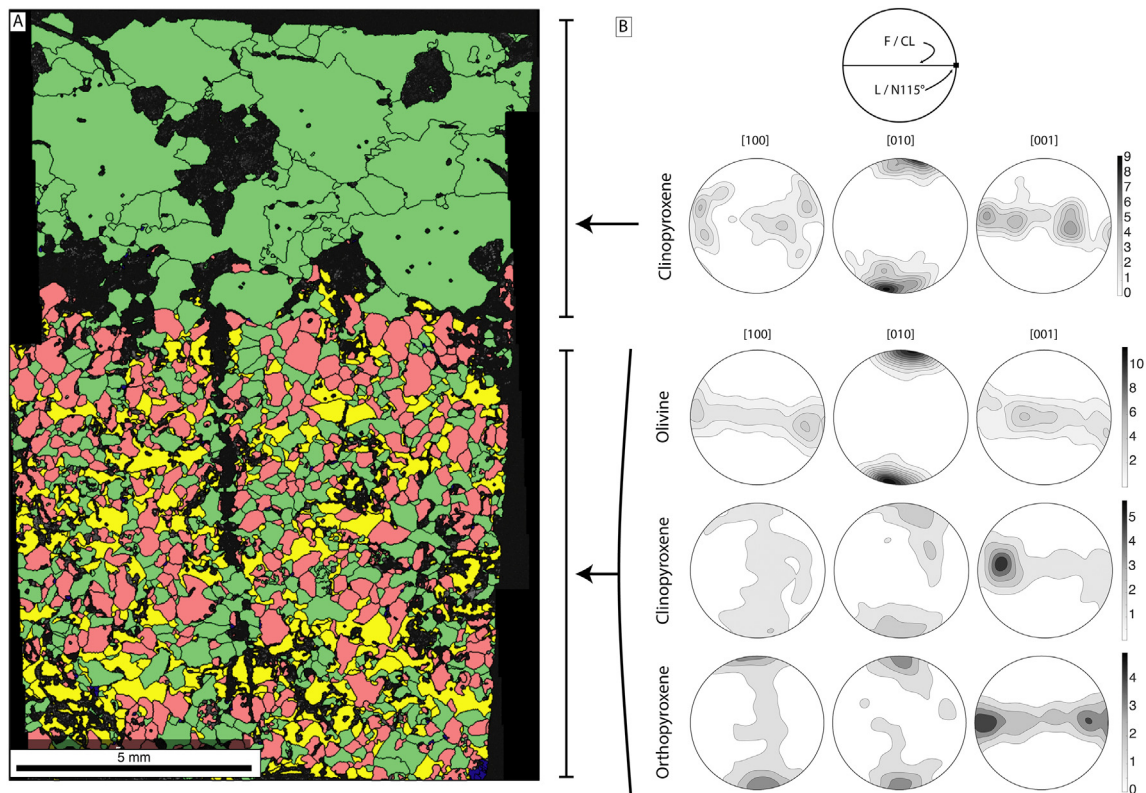


Fig. 8. (A) Colour-coded EBSD map showing the phase distribution in a clinopyroxenitic intrusion found within the layering of a finely-layered cumulate (02T117A). Clinopyroxenitic intrusion is the top part while the finely-layered cumulates make up the lower part of the sample. Cpx is in green, opx in red, olivine in yellow and amphibole in blue. (B) CPO of cpx and CPO of cpx, opx and olivine in sample 02T117A. The structural reference frame for all the pole figures shown in this study is presented. F/CL refers to the position of either the foliation or the compositional layering, depending on the sample. L/N115° refers to the position of the lineation or the N115° direction depending on the sample. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(Figs. 3E and 9) was collected less than 2 m below the finely layered cumulates of the Bear Creek intrusion in the cross section. It is the closest lherzolite sample to the contact between the intrusion and the host rock and its olivine fabric can be interpreted as a [001](010) (B-type, Jung and Karato 2001). We suggest that the possible change in the olivine fabrics between A- and B-type could be related to ongoing plastic deformation conditioned by infiltration of melt from the intrusion, which could trigger a fabric switch either by the integration of structural water via diffusion and/or the presence of interstitial melt (Jung and Karato 2001; Kohlstedt and Holtzman 2009; Qi et al. 2018).

7.2. Magmatic and plastic processes recorded in the Bear Creek intrusion

Ceuleneer and Le Sueur (2008), based on the igneous stratigraphy, proposed that the Bear Creek intrusion evolved in a closed system following the injection of a single batch of magma in the shallow lithosphere. Studying the microstructure of cumulates could thus potentially provide valuable information on the dynamics of crystallisation and direction of transport in this specific magmatic environment. Below, we decipher the progression of the deformation in the section.

7.2.1. Constraints from the finely layered cumulates

At the base of the cumulate section, our field observations revealed that the finely layered cumulates display a well-defined planar structure corresponding to the compositional layering (Fig. 2). On the layering plane, no lineation or foliation (i.e. crystal shape-preferred orientation) were visible at the outcrop or hand specimen scale. In the finely layered cumulate microstructure, we observed the occurrence of growth twins in cpx. The quantification of mineral SPOs shows that the long axis of

the larger population of grains of olivine and cpx are subparallel to N°115, while opx long axis are preferentially aligned at about 90 degrees to cpx and olivine long axis (i.e. subparallel to the layering pole, Fig. 6A). The orthogonal shape-preferred orientation of opx relative to the orientation of cpx and olivine (note the slight elongation of opx in Fig. 4B), and very scarce internal deformation (Fig. 5A) are not consistent with extensive plastic deformation. All the above suggest that the macro- and microstructures of the finely layered cumulates were not overprinted by extensive plastic deformation and probably preserve magmatic features. Regarding the mineral fabrics, olivine, opx and cpx CPOs have [001] axes forming girdles on the compositional layering. This has been observed in a magmatic context (e.g. Holness et al. 2017) and is consistent with the preservation of magmatic features in the finely layered cumulates that was hinted at by field and microstructural observations.

Although no macroscopic lineation was detected, opx and cpx [001] together with olivine [001] or [100] crystallographic axes define submaxima aligned along the regional N°115 orientation, which is also the stretching lineation in the mantle peridotites (Fig. 11). Le Sueur et al. (1984) and Boudier et al. (1989) interpreted this lineation in the peridotite as reflecting asthenospheric flow channelised parallel to the azimuth of the former spreading centre. The olivine and pyroxene clusters can have two origins. (1) Formed through mechanical alignment of crystals by rotation around the vorticity axis of minerals during magmatic flow, when the proportion of melt is still high or (2) formed by limited plastic deformation at low melt volume probably driven by the regional stress field in the case of the Bear Creek intrusion (Boudier et al. 1989; Le Sueur et al. 1984). The limited evidence of plastic deformation reported here, such as the few subgrains and undulose extinction in olivine, cpx and opx (Fig. 5A) could be linked to plastic

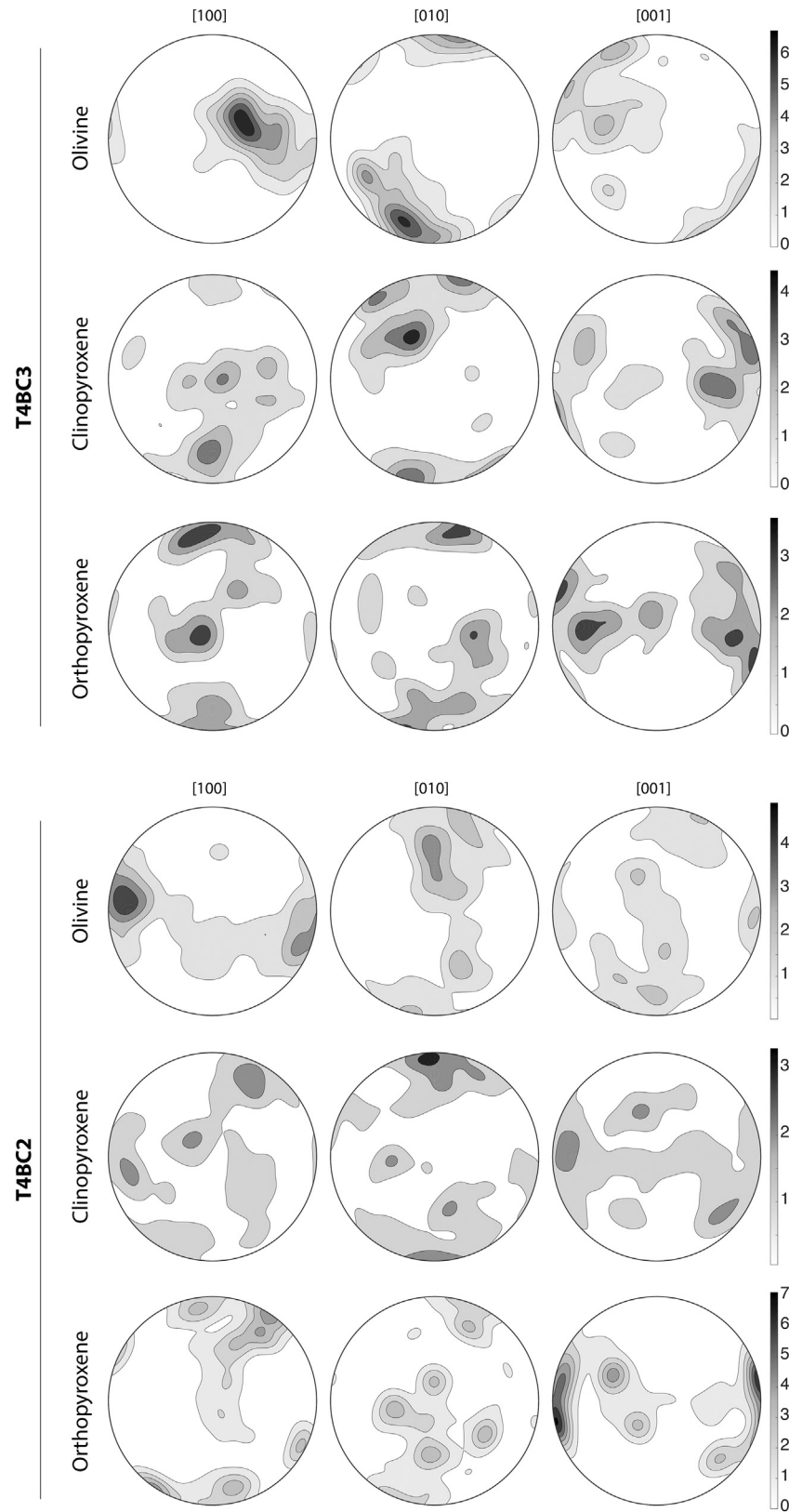


Fig. 9. Crystallographic preferred orientation of cpx, opx and olivine in the host lherzolite from Bear Creek. Additional statistics on the fabrics are available in Supplementary Material A.

deformation under low stress (i.e. regional stress) or alternatively could be the result of the compaction process.

We suggest that the texture and CPO observed in the finely layered cumulates may have been formed by the following processes.

(1) Rock-forming minerals crystallising with a shape anisotropy would be deposited and a preferred orientation was induced by a moderate magmatic flow; (2) compaction of the cumulates leading to limited plastic deformation and to the accentuation of the planar feature

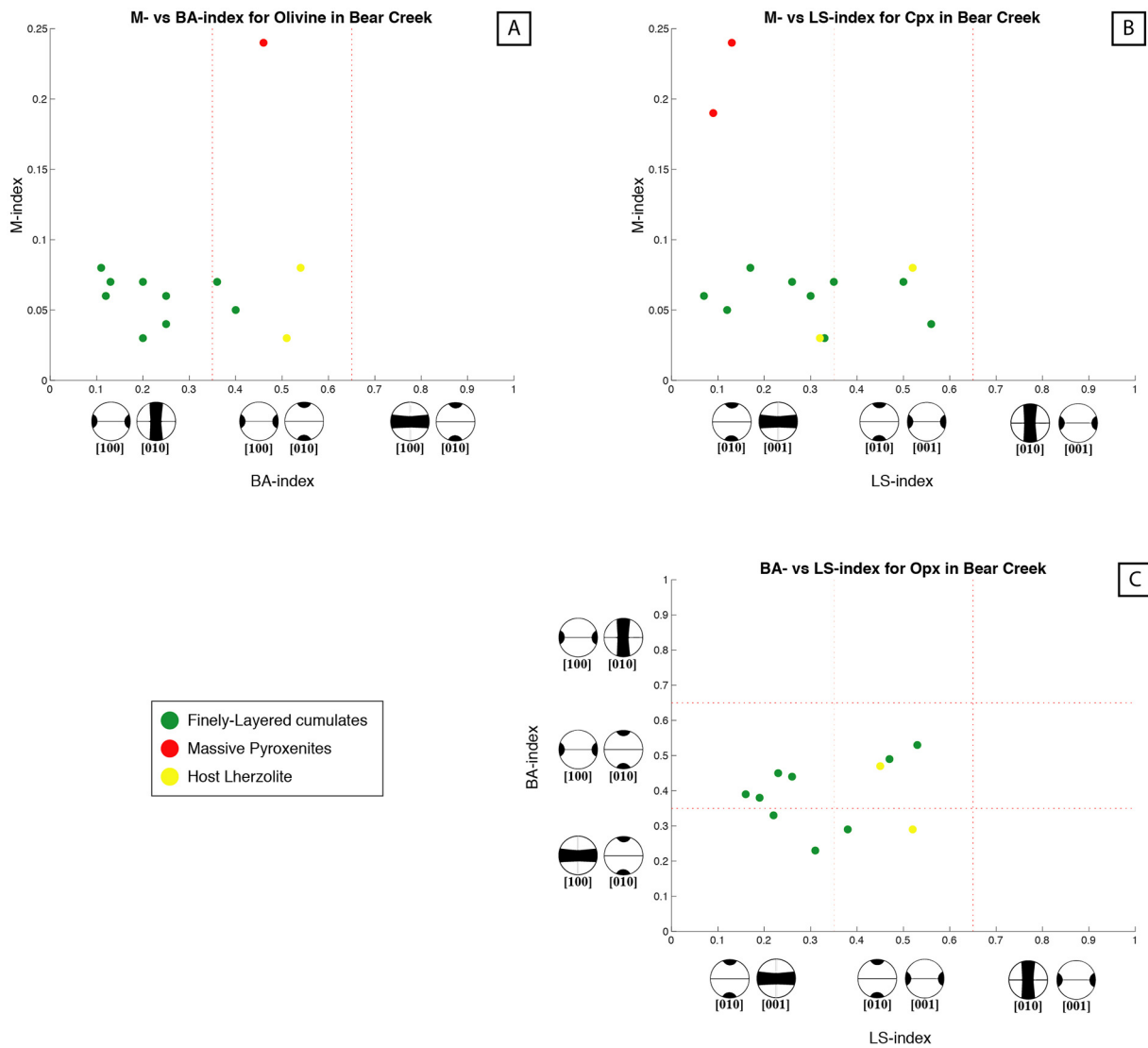


Fig. 10. Fabric symmetry indices of the Bear Creek samples. BA-index and LS-index of olivine and cpx, respectively, are plotted against their respective M-index. BA- and LS-indices of opx were plotted against each other. Finely-layered cumulates are plotted in green, massive pyroxenites in red and lherzolites and peridotitic enclaves in yellow. Vertical dashed red lines at 0.35 and 0.65 mark the limit between the [010]-axial / S-type (lower than 0.35), orthorhombic symmetry and [100]-fibre / L-type (greater than 0.65) domains. Sketches of pole figures represent the ideal pole figure for the different ranges of BA- and LS-indices. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

together with a reduction of the linear feature (Figs. 2A, 4, 6 and 11). Alternatively, an additional and limited episode of deformation could have occurred and could have been responsible for the weak clusters of [001] in pyroxenes and olivine contained in the girdles.

If an initial shape anisotropy was present during crystal settling, our results suggest that cpx crystals are likely to settle mainly on the (010) plane, and to a minor extent on the (100) plane. In EBSD-derived orientation maps, we observe two populations of opx SPO. The first population displays a long axis contained in the compositional plane and has the [010] axis parallel to the layering plane pole. The second population has a long axis perpendicular to the layering plane and has the [001] axis parallel to the layering pole (Supplementary Material B). Based on these observations, we suggest that (010) is more likely to be the settling plane for opx than (100) as opx grains with a long axis contained within the compositional-layering plane preferentially have their [010] axis normal to the layering plane. In Bear Creek, olivine crystals tended to preferentially settle on the (010) plane. For all three minerals, the [001] axis commonly is within the layering plane, consistent with the fact that this axis coincides with the direction of crystal elongation.

These results are consistent with previously-reported fabrics in layered intrusions from the crustal section of the Oman ophiolite. (Benn and Allard 1989) showed that the (010) planes of olivine and cpx coincide with the compositional layering. Similar observations were made in recent studies of layered intrusions from different geological settings (e.g. Holness et al. 2012; Vukmanovic et al. 2018). Our observations are also consistent with recent modelling and the study of adcumulates from a mafic-ultramafic intrusion in the Tarim Craton, which showed that compaction can lead to weak B-type fabrics and formation of subgrains in olivine (Yao et al. 2019).

7.2.2. Constraints from the massive cumulates

In a previous study of the Bear Creek intrusion, Ceuleneer and Le Sueur (2008) used the continuous gradational lithological change and the continuous and correlated variations in major elements (i.e. Mg#) to interpret the finely layered cumulates and the massive cumulates as the crystallisation product of a single crystallisation sequence. The massive cumulates are essentially mono- to bi-mineralic in the olivine-poor level, and exhibit a coarser grain size and straighter grain

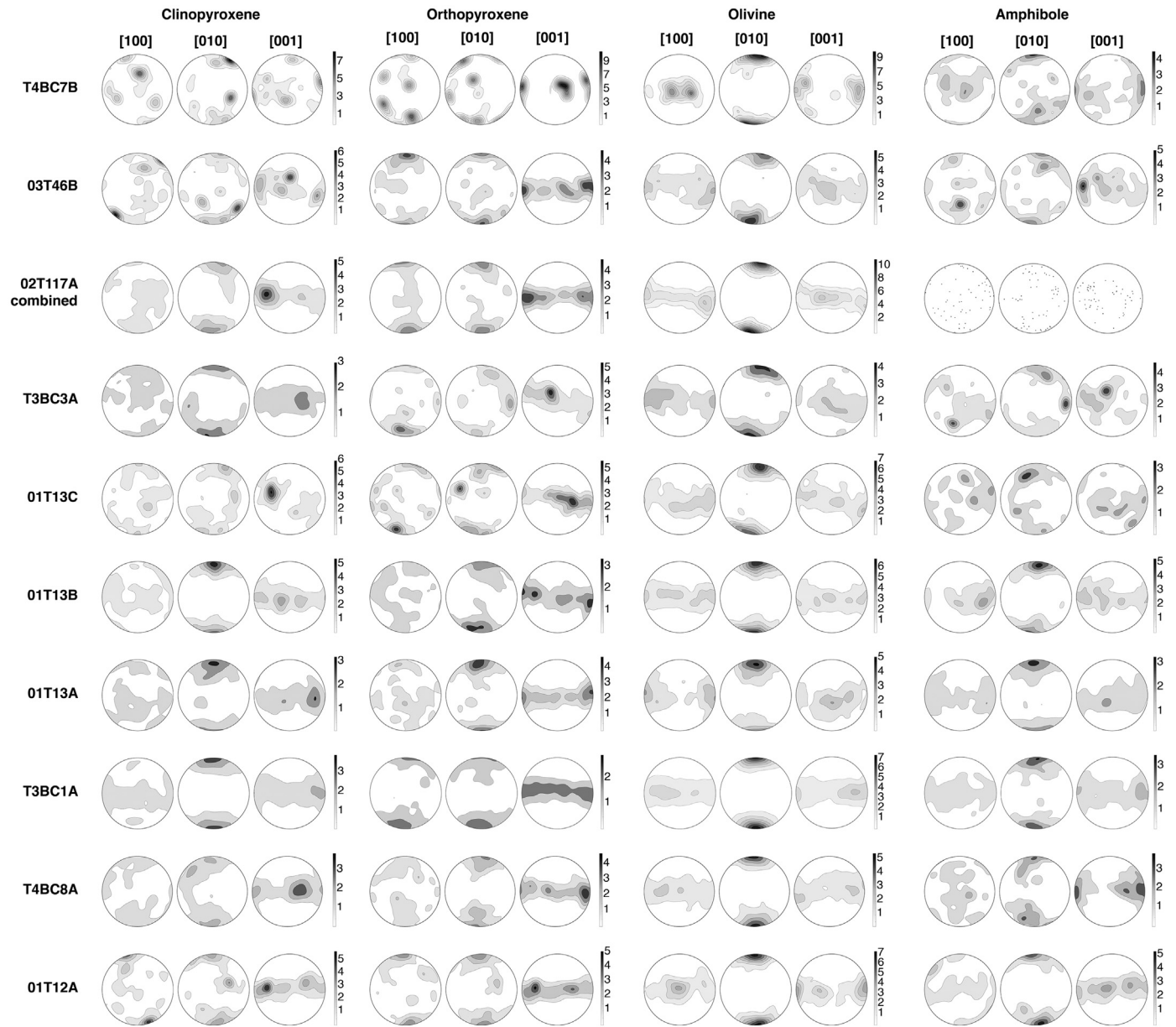


Fig. 11. Pole figures of olivine, cpx, opx and amphibole for the finely layered cumulates analysed in this study. For each phase, the axes [100], [010] and [001] are plotted. Further statistical values can be found in Supplementary Material A.

boundaries than the finely layered cumulates (Figs. 3D and 5B). Our results show that the fabrics observed in the massive cumulates, with development of girdles on pyroxene [001] axes, are comparable to those of their finely layered counterparts (Fig. 12). The quantification of cpx SPO shows a range of orientations from sub-parallel to the olivine and cpx in the finely layered cumulates (parallel to the lineation) to a maximum at small angle to the lineation (Fig. 6). This fan of orientations could be due to the nearly monomineralic composition of the layer triggering further crystal growth.

The lack of deformation markers observed during the field investigation, the relatively scarce internal deformation of large cpx crystals (Fig. 5B), and the CPO of cpx showing a girdle on [001] with weak clusters (Fig. 12) mimic what was observed in the finely layered cumulates (Fig. 5A and 11). Considering the above and the concentric subgrain boundaries observed in the cpx in the massive cumulates, our data suggest that the microstructure of the massive cumulates is mostly magmatic. A similar combination of processes was at play in the finely

layered cumulates. This includes a weak magmatic flow and a limited episode of plastic deformation (probably due to compaction or to regional stress). In this case, the coarser cpx grain size and straighter grain boundaries in the massive pyroxenites can be interpreted as a combined consequence of the rock composition and the crystallisation processes. The crystallisation of a single liquidus phase would favour growth of early grains (rather than high nucleation rate), possibly by grain-boundary migration at high temperature, as suggested by the complex grain-boundary shapes of some grains in the massive pyroxenites (e.g. Fig. 5B). Conversely, in the finely layered cumulates, the homogeneous distribution of the phases will result in a smaller grain-size and can satisfactorily account for the curvilinear grain boundaries at the contact between different phases (Figs. 3 and 5). The fabric of olivine in the massive cumulates plots into the orthorhombic symmetry (Fig. 10) and contrasts with that of cpx. Based on the difference in fabric symmetry and the likely preservation of magmatic microstructures in the cpx-rich layers, we suggest that the olivine-rich layers

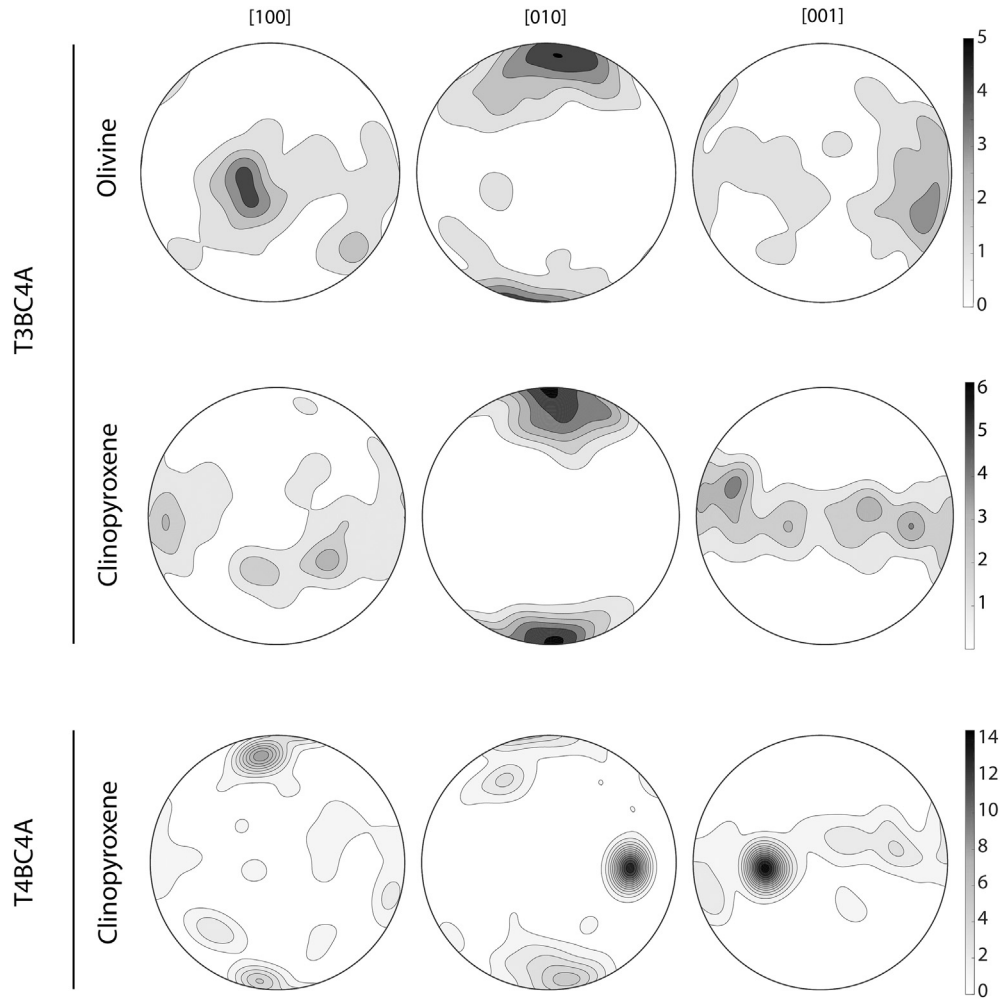


Fig. 12. (A) CPO of olivine and cpx and (B) colour-coded EBSD map of Bear Creek massive cumulate (T3BC4A) showing the misorientation relative to the mean grain orientation for cpx. The larger grain size and the lattice misorientation are visible as well as the equilibrated triple grain junctions. The strong point maximum for cpx [010] and [001] of sample T4BC4A is due to a twinned cpx megacryst.

may have preferentially recorded and accommodated any deformation subsequent to the magmatic stage (i.e. compaction or solid state deformation).

7.2.3. Constraints from the clinopyroxenitic intrusions

Clinopyroxenite intrusions cross-cut the layered pyroxenitic cumulates (Fig. 3A and C), and have a pegmatitic texture (i.e. grain size from 1 mm to multiple centimetres). The grain size of the clinopyroxenitic intrusions can decrease towards the rims of the dykes and may approach the grain size of the host cumulate layers. This can make such intrusions difficult to distinguish from the massive pyroxenites. The presence of numerous cpx twins (Fig. 7B and C) and the low internal deformation suggest a purely magmatic origin for their microstructure. However, and contrary to the finely layered cumulates and massive cumulates, the relatively small size (~ 5–10 cm) of the clinopyroxenitic intrusions does not allow compaction and crystal settling to operate. The textures and fabrics observed in the sample of a clinopyroxenitic intrusion that preserves the contact with finely layered cumulates (02 T117–1; Fig. 8A and B) are indistinguishable from those observed in the massive pyroxenites (Fig. 5B and 12). In the finer-grained part of the clinopyroxenitic intrusions (i.e. part shown in Fig. 8), epitaxial crystallisation of new cpx grains on the cpx already present in the finely layered cumulates or massive pyroxenites can explain the development of the observed fabric. In the pegmatitic parts

of the clinopyroxenitic intrusions (sample 03T16B, Fig. 7B, C and Supplementary Material E), the orthorhombic symmetry of the cpx fabric could be interpreted as a clue to a magmatic flow geometry within the dyke, which would not be purely linear. However, the presence of twinned cpx leads to an over-sampling of the large grains during data collection, thus creating an impression of orthorhombic symmetry while the actual symmetry could be linear (which would be expected within a dyke). Although further investigation is required (in particular, analysis of their respective chemical compositions), it cannot be excluded that some intrusions and massive pyroxenites could be cogenetic, especially in sections where no olivine-rich levels are visible.

7.2.4. Amphibole in the Bear Creek

Small amounts of igneous amphiboles (brown hornblende and pargasite) occur in every lithology from the Bear Creek area (Table 1). Amphibole is usually found as interstitial grains (Fig. 4), is commonly associated with cpx and can also appear as veins discordant to the layering (01T13A, Fig. 4C). In our samples, the amphibole CPO systematically mimics that of the cpx (i.e. [001](010)). This fabric contrasts with the previously-reported dislocation-creep-induced fabrics in amphibole (e.g. Ko and Jung 2015). The observation that an amphibole-rich vein cross-cutting the pre-existing layering (Fig. 4C) shows a CPO similar to that of disseminated amphiboles within the sample (Fig. 11 and Supplementary Material E) is an indication of a possible melt/rock

or fluid/rock-interaction origin for the amphiboles in Trinity. The textural relationship between cpx and amphibole, together with the mimicking of the crystallographic preferred orientations of cpx by the amphibole, also point to a possible epitaxial growth of the amphibole as already observed in other localities (e.g. Henry et al. 2017; Kaczmarek et al. 2016; Moghadam et al. 2020).

7.3. A scenario for the emplacement of the Bear Creek intrusion

The magmatic processes and the plastic deformation recorded in the Bear Creek intrusion share similarities with other intrusions emplaced in extensional settings (e.g. Holness et al. 2012; Holness et al. 2017; Vukmanovic et al. 2018 and reference therein). However, the Bear Creek intrusion presents some features that could be typical for this type of environment, and lead us to draw a scenario for its emplacement into the mantle domain exposed in the ophiolites.

- (1) An episode of solid-state flow occurred in ambient Trinity mantle wall-rock, mostly under high-temperature and low-stress conditions in the dislocation creep regime. This episode probably occurred as the “asthenospheric” mantle rose in an oceanic spreading environment and was channelised at shallow depth along the spreading axis (e.g. Boudier et al. 1989; Dygert et al. 2016; Le Sueur et al. 1984). Pressure probably was below 1 GPa (Quick 1981) and the temperature above 1000 °C so that an A-type olivine fabric could form as observed in the Iherzolite.
- (2) In the Bear Creek area, intrusion of a boninitic magma at shallow depth (<1 GPa) led to the crystallisation of opx, cpx and olivine, producing finely layered cumulates, then with further differentiation, massive pyroxenites dominated by cpx (Ceuleneer and Le Sueur 2008). During that accumulation of ultramafic minerals, a dominantly planar mineral fabric developed in the cumulates due to the combination of a crystal settling with weak magmatic flux. With decreasing melt proportions, compaction probably operated; it could have accentuated the planar component of the lithologies and formed the subgrains observed in the minerals of the Bear Creek samples. In the fabrics, we observe the presence of sub-maxima of preferred orientation for cpx, opx or olivine crystallographic axes within girdles coinciding with the compositional layering (Fig. 9). This may provide constraints on the geometry of the magmatic flow in a magma-chamber setting, and/or it may reflect the persistence of the regional stress field. Our observations suggest that the magmatic processes recorded are homogeneous over the entire section. Later intrusions occurred, crystallising into coarse-grained to pegmatitic clinopyroxenite-rich veins and dykes crosscutting the weakly-consolidated cumulate layers. Finally, the crystallisation of amphibole was followed by epitaxial growth on cpx.

8. Conclusions

New field and EBSD data have allowed us to characterise the deformation mechanisms and the formation/deformation history of the Bear Creek layered pyroxenitic cumulates, and their relationships with the host mantle peridotites (Iherzolite) in the Trinity ophiolite. This detailed microstructural study of the Bear Creek intrusion suggests a scenario for the formation of layered intrusions that first includes crystal settling, then magma flow and later compaction. CPO, SPO and internal deformation of minerals suggest that crystal settling is probably the major process creating the anisotropy of the cumulates, rather than magma flow, crystal growth or compaction. Our data show that plastic deformation is present but not widely recorded in any minerals, even at the base of the section. We interpret this as indicating a limited contribution of compaction or alternatively for an ongoing regional stress to the microstructure. From a purely microstructural aspect, this study emphasises

the importance of crystal-shape control during the development of microstructures in cumulate rocks.

In the Trinity complex, magmatic processes gave rise to strong crystallographic preferred orientations for olivine, cpx and opx, similar to those classically attributed to plastic flow in the dislocation creep regime. Since these are likely to constrain the activation of favourably oriented slip systems during subsequent deformation, the reliability of their use as proxies for the conditions of deformation has to be regarded with caution.

Our results, along with those of previous structural and geochemical work, are consistent with (1) the mantle section of Trinity ophiolite representing a fossil marginal slow-spreading centre obducted onto the North American plate and (2) Bear Creek and similar intrusions corresponding to the late- to post-tectonic emplacement and differentiation of boninitic magmas in a subduction-related environment.

The microstructural data that we have presented may also have important implications for our understanding of the development of the fabrics in pyroxene-rich lithologies, which could be pre-conditioned by the initial magmatic fabric. The strong fabrics in the lithologies in the Bear Creek cumulates would also have important implications in terms of seismic anisotropy as they differ significantly from those of more classic mantle pyroxenites.

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Declaration of Competing Interest

None.

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